Md Abu Sayeb Rayhan

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Google Scholar | ResearchGate | GitHub | LinkedIn

PROFILE

2+ years in Japan and Bangladesh, including real-time retail video analytics (tracking, pose, activity recognition) on IoT edge devices; **seven papers, five first-authored**, in medical AI and robot perception, including preregistered studies in 200 homes. Interests: **trustworthy vision**, **robot perception**, **edge AI**.

EDUCATION

Bachelor of Science, Computer Science & Engineering GPA: 3.4 / 4.0 Jan 2019 – May 2024

Comilla University, Bangladesh

Relevant Coursework: Artificial Intelligence, Algorithm Design, Theory of Computation, Introduction to Data Science, Object-Oriented Programming, Research-Based Project.

Bachelor's Thesis: *Text-to-Image Generation using **Diffusion Models***. Fine-tuned Stable Diffusion (U-Net backbone) on a custom dataset of Bangladeshi landmarks. Investigated prompt engineering and conditioning mechanisms for generative fidelity.

PUBLICATIONS

Seven papers, five as first author (my name in bold). Abstracts, figures and slides: abusayeb.github.io/research

Peer-reviewed conference papers

[1] **M. A. S. Rayhan**, M. Fahimah, M. T. Mahmud, H. Hiya, A. S. Dhar. "RGEF: Reliability-Gated Evidential Fusion for Robust Medical AI." *IEEE Int. Conf. on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, 2026. *In press.* [paper page]

[2] **M. A. S. Rayhan**, A. Shutradhar, H. Hiya. "Forensic Invariant Learning for Synthetic Image Detection: Bridging Benford's Law and Topological Analysis with Machine Learning Ensembles." *IEEE Int. Conf. on Quantum Photonics, Artificial Intelligence and Networking (QPAIN)*, 2026. [IEEE Xplore]

[3] A. Shutradhar, **M. A. S. Rayhan**, M. Khatun, B. Das. "BD Eye-Doc: When Computer Vision Meets Conversational AI for Scalable Eye Disease Screening in Bangladesh." *IEEE QPAIN*, 2026. [IEEE Xplore]

[4] A. S. Dhar, B. Das, M. M. Hassan, **M. A. S. Rayhan**, H. T. Saydia, S. A. Shimu. "Is CNN Dead? A Comprehensive Empirical Analysis of Convolutional Neural Networks, Ensemble Methods, and Vision Transformers for Rice Leaf Disease Classification in Low-Resource Agricultural Settings." *Int. Conf. on Emerging Computing and Communication Technologies (ECCT)*, 2026; Atlantis Press (Springer Nature). *In press.*

Journal manuscripts submitted

[5] **M. A. S. Rayhan**, H. Hiya, A. S. Dhar. "Does learning when to look pay for home service robots? A preregistered test of adaptive perception scheduling in closed-loop navigation." Submitted to *Intelligent Service Robotics* (Springer), 2026. [preprint]

[6] **M. A. S. Rayhan**, H. Hiya, A. S. Dhar. "Doorways set the pace of robot vacuum cleaning: a parameter-free room-graph model, tested and confirmed in 200 real homes." Submitted to *Robotics and Autonomous Systems* (Elsevier), 2026.

[7] **M. A. S. Rayhan**, H. Hiya, A. S. Dhar. "Input-conditional uncertainty fusion for skin lesion screening." Submitted to *Biomedical Physics & Engineering Express*, 2026.

RESEARCH EXPERIENCE

Independent Researcher – Dhaka, Bangladesh Apr 2026 – present

With H. Hiya (Jahangirnagar University) and A. S. Dhar (Comilla University)

- Designed and ran a **preregistered closed-loop study** of adaptive perception for home robots [5] in 25 held-out scanned homes (Habitat, HM3D) against 12 baselines; the learned scheduler cut detector compute by 85% but missed its preregistered success and safety bar, and the paper reports that negative result.
- Derived and tested a **parameter-free room-graph model** of robot-vacuum cleaning [6] over 17,580 simulated runs in 200 homes; a second preregistration on 100 unseen homes confirmed it ($R^2 = 0.81$).
- Extended RGEF into a journal study [7]: per-image reliability gating of eight uncertainty signals on a deduplicated external archive raised AUROC from 0.6945 to 0.7946.

Research Assistant – Dept. of CSE, Comilla University Jan 2023 – Mar 2024

Supervisor: Partha Chakraborty

- Designed datasets, ran experiments and analysed results for empirical studies in human–robot interaction and optical character recognition; results were carried into peer-reviewed conference papers.

PROFESSIONAL EXPERIENCE

Data Scientist – Othello Financial Services Co. Ltd., Osaka, Japan Oct 2024 – Mar 2026

AGiFT AI Camera team

- Designed and deployed multi-object tracking (ByteTrack), pose estimation (MoveNet), and action recognition (MMAction2) pipelines on resource-constrained **edge devices** for real-time retail video analytics.
- Optimized and quantized vision models using **TensorRT / ONNX Runtime** for low-power edge inference, and parallelized execution to cut processing latency by **about 70%**.
- Fine-tuned a YOLO detector (90%+ accuracy) for edge-assisted event segmentation that automated sports highlight extraction, cutting production cost by **50%**; generalized pipeline to live crowd streams at festivals and concerts.
- Integrated in-store edge devices with cloud infrastructures on Microsoft Azure, deploying a retrieval-augmented generation system (LangChain, Pinecone, GPT-4) connected to real-time IoT vision alerts.

AI Engineer (contract) – Government of Bangladesh project, Cumilla Feb 2024 – Jul 2024

- Led a three-person team that delivered a bilingual (Bangla/English) document-search system combining OCR with dense embeddings across PDF, Word, Excel and image files, shipped as an offline desktop application (PyQt, ChromaDB).

Founder and developer – eGP AutoPilot, Dhaka 2026 – present

Alongside independent research

- Designed, built and released a bilingual desktop application that automates tender submission on Bangladesh's national e-Government Procurement portal (Playwright, PyQt, Firebase); v2.2.0 ships on macOS and Windows with 273 offline tests.

TECHNICAL SKILLS

- **Programming:** Python (advanced), C/C++, SQL, Java
- **ML / Edge AI Frameworks:** PyTorch, TensorFlow, TensorRT, ONNX Runtime, OpenCV, MMAction2, YOLO, ByteTrack, MoveNet, Stable Diffusion, LangChain, Hugging Face, Scikit-learn
- **Methods:** Edge AI Deployment, Model Quantization, Object Detection, Action Recognition, Pose Estimation, Spatial Clustering, Diffusion Models, OCR, RAG
- **Infrastructure:** Edge IoT Devices, Microsoft Azure, Docker, CI/CD, Git; Flask, PyQt; Pinecone, Azure SQL

LANGUAGE PROFICIENCY

English – IELTS 7.0 (Listening 7.5, Reading 8.0, Writing 6.5, Speaking 6.5) | Bangla – Native | Japanese – Basic Daily Conversation (JLPT N4 equivalent)

LEADERSHIP & SERVICE

- **Team lead**, Government of Bangladesh project (2024): led a team of three from design to delivery.
- **Joint General Secretary**, North Bengal Student Association, Comilla University (2023–2024): elected; organised events and student networking and took part in policy discussions.
- **Batch Representative**, Dept. of CSE, Comilla University (2022–2024): elected by peers to represent the cohort to faculty and administration on academic quality and student welfare.
- Took part in student-led initiatives promoting technology awareness and collaborative learning at Comilla University (2019–2024).

HONOURS, AWARDS & PRESENTATIONS

- **Best Presenter Award**, IEEE QPAIN 2026.
- Conference presentations: IEEE RAAICON 2026 (RGEF, slides); IEEE QPAIN 2026; ECCT 2026.
- Inter-University Programming Contest, Comilla University (2019–2021); certified in Mobile Game & Application Development, ICT Division, Government of Bangladesh (2022).
- Inter-department sports, Comilla University (2019–2023): cricket champions; third place in volleyball.

REFERENCES

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